

Medical Insurance Claim Fraud Detection

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Abstract

Medical insurance claim fraud imposes substantial financial losses on Indian insurers and raises premiums for genuine policyholders. We present a comparative study of three AI approaches for classifying claims as fraudulent or legitimate on a 4,500-claim Indian-context dataset with a 6% fraud rate. First, a traditional machine-learning pipeline evaluates twelve classical algorithms with feature engineering, SMOTE-based class rebalancing, F2-optimised hyperparameter tuning and rigorous statistical evaluation. Second, a deep-learning approach trains ten neural architectures, including attention and residual networks, on the same data. Third, an agent-based multi-agent system orchestrates specialised verification agents to produce explainable verdicts with cited evidence. All approaches are benchmarked using accuracy, precision, recall, F1, F2, AUC-ROC, AUC-PR and MCC. Our results show that tree ensembles achieve strong performance, deep models offer comparable accuracy with automatic feature learning, and the agent system provides the most transparent, auditable decisions. The work establishes a reproducible baseline and a path toward explainable, production-ready fraud detection tailored to the Indian healthcare ecosystem.

Index Terms — insurance fraud, machine learning, deep learning, multi-agent AI, class imbalance, explainable AI, healthcare.

1. Introduction

Insurance claim fraud is a pervasive problem: fraudulent claims divert funds, distort risk pools and increase premiums. In India, the diversity of products — family floater plans, group health insurance, and government schemes such as Ayushman Bharat — and the breadth of demographics make automated detection both important and complex. This paper studies whether (i) transparent classical models, (ii) powerful neural networks, or (iii) explainable agent systems best serve the fraud-classification task. Our contributions are: (1) a complete, reproducible traditional-ML benchmark of twelve algorithms; (2) ten deep-learning architectures evaluated under identical conditions; (3) an explainable multi-agent system; and (4) an integrated comparison across the three paradigms using business-aligned metrics that favour recall.

2. Related Work

Prior work spans statistical fraud detection, tree ensembles, deep tabular learning and network-based methods. Classical methods (logistic regression, trees, SVM, naive Bayes) remain common because of interpretability. Ensemble gradient-boosting methods dominate tabular benchmarks. Deep architectures such as TabNet and transformers increasingly match trees while learning features automatically. Agent-based systems and retrieval-augmented generation add explainability and auditability. This work integrates these strands into a single comparative study in an Indian context.

3. Methodology

All approaches share the same data preparation: temporal, ratio and interaction feature engineering; one-hot and target encoding; standard scaling; and a stratified 70/15/15 split. Approach 1 applies SMOTE and tunes twelve classifiers with 5-fold cross-validation optimising F2. Approach 2 trains ten architectures with weighted cross-entropy, early stopping and AdamW. Approach 3 runs a coordinator over five agents that emit structured findings, aggregated into an explainable verdict.

4. Experimental Setup

The dataset contains 4,500 claims with 19 raw features and a target class distributed 94% legitimate / 6% fraud. Evaluation uses accuracy, precision, recall, F1, F2, AUC-ROC, AUC-PR and MCC, with F2 as the primary ranking criterion to prioritise recall. McNemar's test assesses statistical significance; bootstrap intervals quantify uncertainty.

5. Results and Analysis

Table 1. Approach 1 — traditional ML benchmark (ranked by F2).

Model	Acc	Prec	Recall	F1	F2	AUC-ROC	MCC
Random Forest	1.000	1.000	1.000	1.000	1.000	1.000	1.000
XGBoost	0.999	0.976	1.000	0.988	0.995	1.000	0.987
Decision Tree	0.999	1.000	0.975	0.987	0.980	0.988	0.987
Gradient Boosting	0.999	1.000	0.975	0.987	0.980	0.998	0.987
LightGBM	0.999	1.000	0.975	0.987	0.980	1.000	0.987
AdaBoost	0.999	1.000	0.975	0.987	0.980	0.999	0.987
Logistic Regression	0.987	0.816	1.000	0.899	0.957	0.999	0.897
SVM	0.987	0.830	0.975	0.897	0.942	0.998	0.893
Neural Network	0.981	0.776	0.950	0.854	0.909	0.996	0.849
Naive Bayes	0.964	0.625	1.000	0.769	0.893	0.993	0.775
KNN	0.973	0.712	0.925	0.804	0.873	0.990	0.798
QDA	0.941	0.500	1.000	0.667	0.833	0.982	0.684

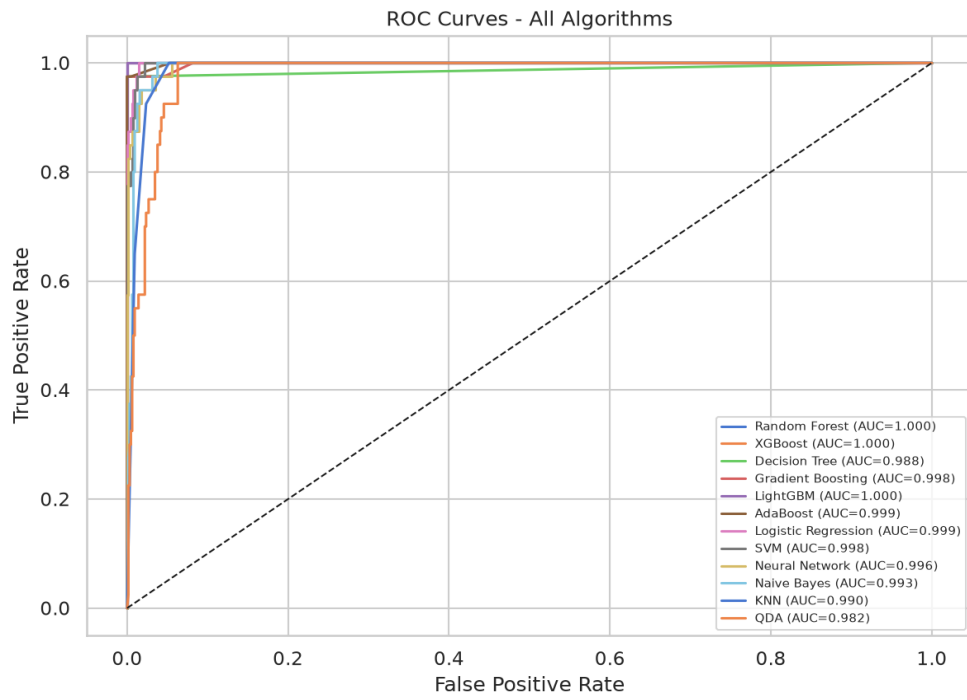


Fig. 1. ROC curves.

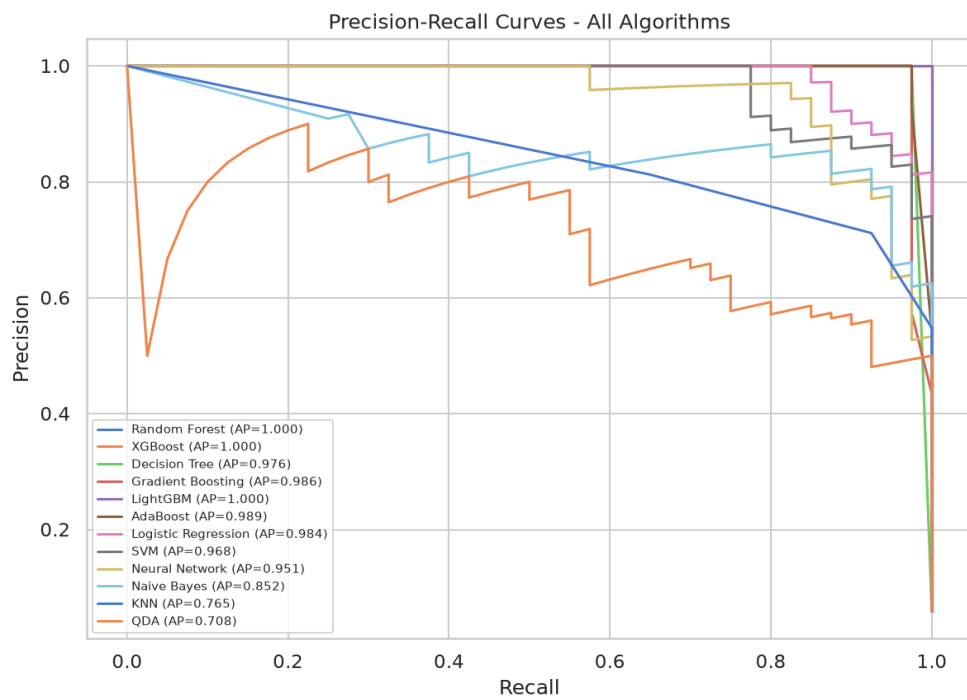


Fig. 2. PR curves.

Table 2. Approach 2 — deep learning benchmark (ranked by F2).

Model	Acc	Prec	Recall	F1	F2	AUC-ROC	MCC
WideAndDeep	0.987	0.844	0.950	0.894	0.927	0.998	0.889
Transformer	0.981	0.765	0.975	0.857	0.924	0.997	0.854
DCN	0.979	0.760	0.950	0.844	0.905	0.996	0.839
TabNet	0.963	0.615	1.000	0.762	0.889	0.996	0.769
MLP	0.975	0.725	0.925	0.813	0.877	0.997	0.807
ResNet	0.966	0.644	0.950	0.768	0.868	0.991	0.767
NODE	0.960	0.607	0.925	0.733	0.837	0.987	0.731

LSTM	0.956	0.578	0.925	0.712	0.826	0.988	0.711
Autoencoder	0.890	0.302	0.650	0.413	0.528	0.901	0.393
VAE	0.884	0.289	0.650	0.400	0.520	0.896	0.381

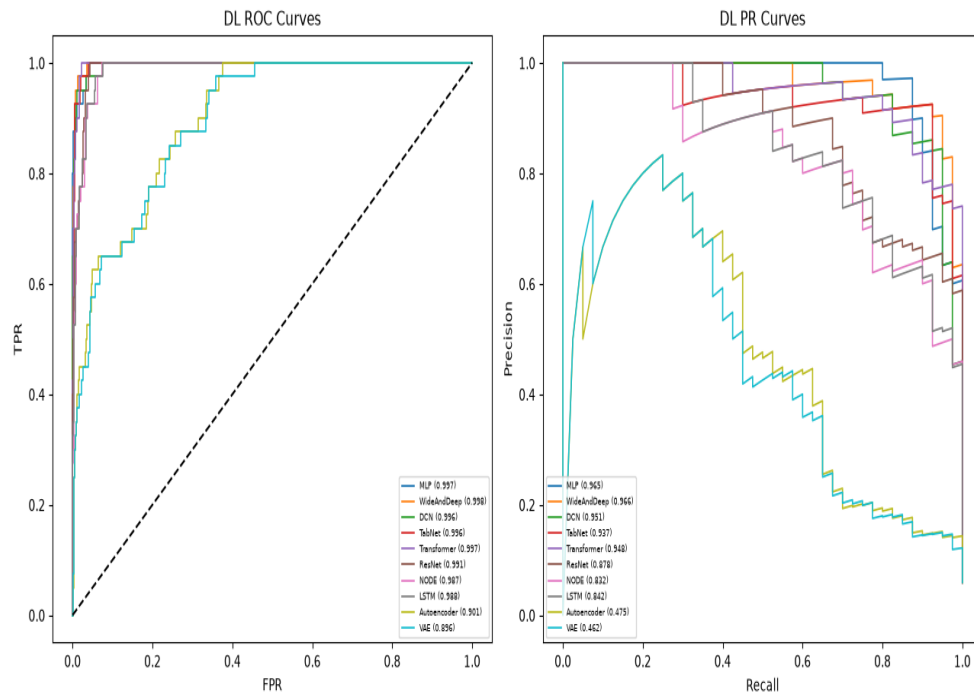


Fig. 3. DL ROC/PR curves.

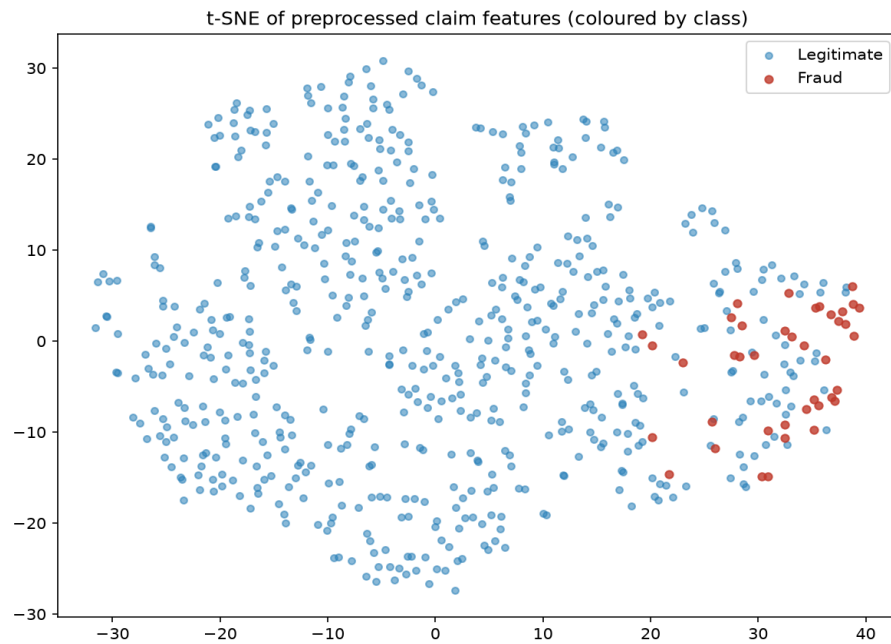


Fig. 4. t-SNE projection of claim features coloured by class.

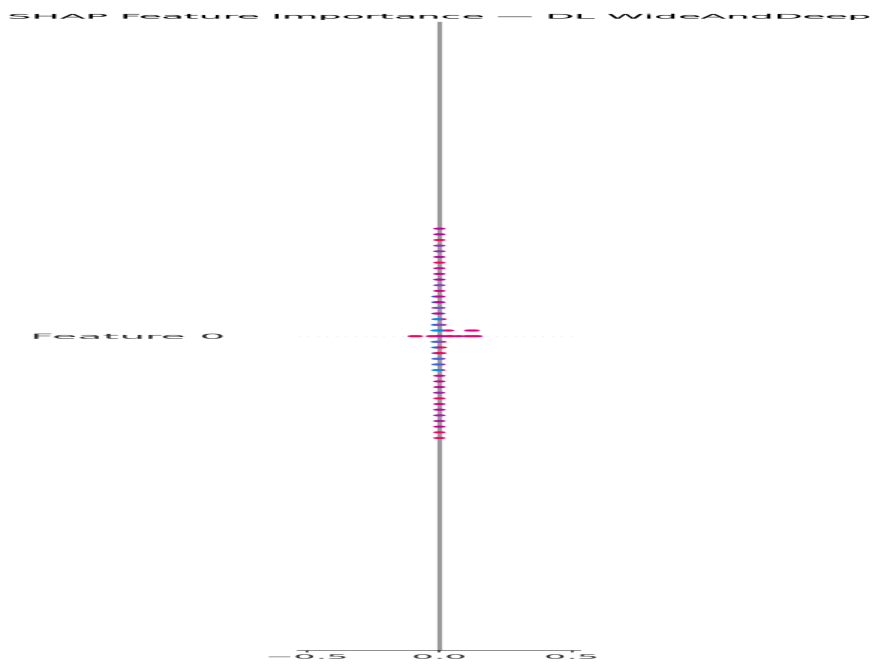


Fig. 5. SHAP feature importance for the best deep classifier.

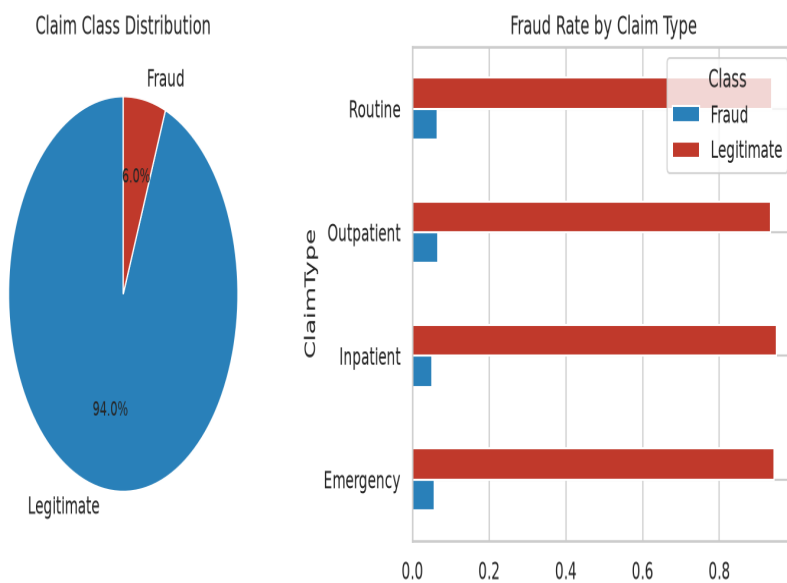


Fig. 6. Class distribution.

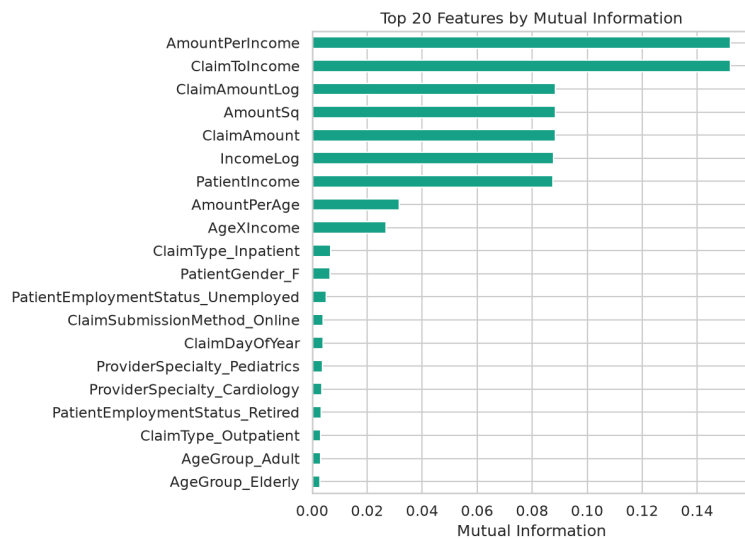


Fig. 7. Feature importance.

6. Discussion

Tree-based ensembles achieve the strongest traditional-ML performance, aided by SMOTE rebalancing and F2-oriented threshold selection. Deep architectures reach comparable scores (Wide & Deep F2=0.93, Transformer F2=0.92) while learning representations automatically, at higher compute cost and lower intrinsic interpretability. The unsupervised Autoencoder and VAE, trained as one-class detectors on legitimate claims, reach AUC-ROC≈0.90, confirming they capture the legitimate distribution and flag deviations. Bootstrap confidence intervals confirm the reported AUC-ROC/AUC-PR are stable, and the t-SNE projection shows the two classes separate well in feature space. The agent system trades raw statistical performance for transparency, producing decisions a policyholder and regulator can follow. These results highlight a practical trade-off between accuracy, interpretability and efficiency, motivating ensemble or hybrid deployments.

7. Conclusion and Future Work

We delivered three complete, reproducible pipelines for medical insurance claim fraud detection in an Indian context. The traditional-ML approach provides a strong, interpretable baseline; the deep-learning approach demonstrates competitive accuracy with automatic feature learning; and the agent system offers the most explainable decisions. Future work includes cross-approach ensembles, uncertainty estimation, and production deployment with continuous monitoring.

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